

Seminar 7

The dimension of a vector space is given by the no. of the vectors in its basis.

$$\begin{aligned} f: A &\rightarrow B \\ \dim(\ker f) + \dim(\operatorname{Im} f) &= \dim A. \end{aligned}$$

The complement of a space A is $\bar{A}$ containing the minimum no. of vectors s.t. $A \oplus \bar{A} = V$ (the whole space)

or to complete A to a basis of V .

For S, T vector spaces:

$$\dim S + \dim T = \dim(S+T) + \dim(S \cap T)$$

$$\ker(f) = \{x \in A \mid f(x) = 0\}, \quad \operatorname{Im}(f) = \{y \in B \mid f(x) = y\}$$

$$A \subseteq B \Rightarrow \dim A \leq \dim B.$$

$$(1) A = \{(x, y, z) \in \mathbb{R}^3 \mid z = 0\}$$

$$B = \{(x, y, z) \in \mathbb{R}^3 \mid x + y + z = 0\}$$

$$C = \{(x, y, z) \in \mathbb{R}^3 \mid x = y = z\}.$$

Det. a basis and dimensions.

$$A = \{(x, y, 0) \in \mathbb{R}^3 \mid x, y \in \mathbb{R}\}$$

$$= \{(x, 0, 0) + (0, y, 0) \mid x, y \in \mathbb{R}\}$$

$$= \{x(1, 0, 0) + y(0, 1, 0) \mid x, y \in \mathbb{R}\}$$

$$= \langle (1, 0, 0), (0, 1, 0) \rangle$$

$$x_1(1, 0, 0) + x_2(0, 1, 0) = (0, 0, 0)$$

$$\begin{aligned} 2) \begin{cases} x_1 = 0 \\ x_2 = 0 \end{cases} &\Rightarrow \text{lin. indep.} \Rightarrow \langle (1, 0, 0), (0, 1, 0) \rangle \text{ basis of } A. \\ &\Rightarrow \dim A = 2. \end{aligned}$$

$$\begin{aligned}
 B &= \{ (x, y, z) \in \mathbb{R}^3 \mid x = -y - z \} \\
 &= \{ (-y - z, y, z) \in \mathbb{R}^3 \mid y, z \in \mathbb{R} \} \\
 &= \{ (-y, y, 0) + (-z, 0, z) \mid y, z \in \mathbb{R} \} \\
 &= \{ y(-1, 1, 0) + z(-1, 0, 1) \mid y, z \in \mathbb{R} \} \\
 &= \langle (-1, 1, 0), (-1, 0, 1) \rangle.
 \end{aligned}$$

← -y? ✓
(there are > basis)

$$\alpha_1(-1, 1, 0) + \alpha_2(-1, 0, 1) = (0, 0, 0)$$

$$\Rightarrow \begin{cases} -\alpha_1 - \alpha_2 = 0 \\ \alpha_1 = 0 \\ \alpha_2 = 0 \end{cases} \Rightarrow \text{lin. independent.} \Rightarrow \langle (-1, 1, 0), (-1, 0, 1) \rangle \text{ basis of } B.$$

$$\dim B = 2$$

$$\begin{aligned}
 C &= \{ (x, x, x) \in \mathbb{R}^3 \mid x \in \mathbb{R} \} \\
 &= \{ x(1, 1, 1) \mid x \in \mathbb{R} \} \\
 &= \langle (1, 1, 1) \rangle \Rightarrow \text{only one vector in the generator} \Rightarrow \text{lin. indep.} \\
 &\quad \Rightarrow \langle (1, 1, 1) \rangle \text{ basis of } C. \\
 &\Rightarrow \dim C = 1.
 \end{aligned}$$

③ Det. a basis and dimensions of the vector spaces $\mathbb{C}$, $\mathbb{R}$

b) Prove that the set containing $\{1, i\}$ lin. dep. in $\mathbb{C}$
lin. indep. in $\mathbb{R}$.

$$\begin{aligned}
 S \neq \emptyset \\
 (\forall) a, b \in \mathbb{R} \\
 (\forall) x, y \in \mathbb{R} \\
 \downarrow \\
 ax + by \in S.
 \end{aligned}$$

$$\begin{cases} \alpha = -i \\ \beta = 1 \end{cases}$$

$$\mathbb{C} \Rightarrow \forall v \in \mathbb{C}, \alpha \in \mathbb{C}$$

$$\mathbb{C} = \langle 1 \rangle \hookrightarrow \text{lin. indep.}$$

Also $(\forall) v \in \mathbb{C}, \exists! \alpha \in \mathbb{C} \text{ s.t. } \alpha \cdot 1 = v$

lin indep + generator $\Rightarrow \langle 1 \rangle$ basis $\mathbb{C} \Rightarrow \dim_{\mathbb{C}} \mathbb{C} = 1$.

$$\alpha_1 \cdot 1 + \alpha_2 \cdot i = 0, \alpha_1, \alpha_2 \in \mathbb{C}.$$

Counterexample: $\alpha_1 = -i, \alpha_2 = 1 \Rightarrow -i \cdot 1 + 1 \cdot i = 0 \Rightarrow \text{lin. dep. in } \mathbb{C}$

$$\mathbb{R} \subset \mathbb{C} : n \in \mathbb{C}, \alpha \in \mathbb{R}.$$

$$\{a+bi \mid a, b \in \mathbb{R}\} = \{a \cdot 1 + b \cdot i \mid a, b \in \mathbb{R}\} = \langle 1, i \rangle.$$

$$\mathbb{R} \subset \mathbb{C} = \langle 1, i \rangle \rightarrow \text{lin. indep. } (\Leftrightarrow \text{not } a \cdot 1 + b \cdot i = 0 \Leftrightarrow a = b = 0)$$

$$(\forall) v \in \mathbb{C}, \exists! \alpha \in \mathbb{R} : \alpha \cdot \langle 1, i \rangle = v.$$

$$\alpha_1 \cdot 1 + \alpha_2 \cdot i = 0$$

$$\Leftrightarrow \alpha_1 = \alpha_2 = 0 \Rightarrow \text{lin. indep.}$$

$$\Rightarrow \langle 1, i \rangle \text{ basis}$$

$$\Rightarrow \dim_{\mathbb{R}} \mathbb{C} = \underline{\underline{2}}!$$

$$(4) f: \mathbb{R}^3 \rightarrow \mathbb{R}^2.$$

$$f(x, y, z) = (y, -x)$$

f is an $\mathbb{R}$ -linear map.

Det. a basis and dimension of $\text{Ker } f$, $\text{Im } f$.

$$f \rightarrow \mathbb{R}\text{-linear map if: } \begin{cases} f(v_1 + v_2) = f(v_1) + f(v_2), (\forall) v_1, v_2 \in \mathbb{R}^3 \\ f(k \cdot v) = k \cdot f(v), k \in \mathbb{R}, v \in \mathbb{R}^3 \end{cases}$$

$$v_1 = (x_1, y_1, z_1)$$

$$v_2 = (x_2, y_2, z_2)$$

$$\begin{aligned} f(v_1 + v_2) &= f((x_1, y_1, z_1) + (x_2, y_2, z_2)) = f(x_1 + x_2, y_1 + y_2, z_1 + z_2) \\ &= f(x_1, y_1, z_1) + f(x_2, y_2, z_2) \\ &= (y_1, -x_1) + (y_2, -x_2) \\ &= (y_1 + y_2, -x_1 - x_2) \end{aligned}$$

$y_1 + y_2, -x_1 - x_2$

True (1)

$$\begin{aligned} f(k \cdot (x_1, y_1, z_1)) &= f(kx_1, ky_1, kz_1) = (ky_1, -kx_1) = \\ &= k(y_1, -x_1) = k \cdot f(v) \quad (2) \end{aligned}$$

(1), (2) $\Rightarrow f$ linear map.

$$\begin{aligned}
 \text{Ker } f &= \{ (x, y, z) \in \mathbb{R}^3 \mid f(x, y, z) = 0 \} \\
 &= \{ (x, y, z) \in \mathbb{R}^3 \mid (y, -x) = 0 \} \\
 &= \{ (0, 0, z) \in \mathbb{R}^3 \mid z \in \mathbb{R} \} \\
 &= \{ z(0, 0, 1) \in \mathbb{R}^3 \mid z \in \mathbb{R} \} \\
 &= \langle (0, 0, 1) \rangle \Rightarrow \text{lin. indep.} \\
 &\quad \text{generates the subspace} \Rightarrow \text{basis.} \Rightarrow \dim(\text{Ker } f) = 1.
 \end{aligned}$$

$$\begin{aligned}
 \text{Im } f &= \{ (x, y, z) \in \mathbb{R}^3 \mid f(x, y, z) = (y, -x) \} \\
 \text{Im } f &= \{ (y, -x) \in \mathbb{R}^2 \mid f(x, y, z) = (y, -x) \} \\
 &= \{ (y, -x) \in \mathbb{R}^2 \} \\
 &= \{ y(1, 0) + x(0, -1) \mid x, y \in \mathbb{R} \} \\
 &= \langle (1, 0), (0, -1) \rangle
 \end{aligned}$$

$$\alpha(1, 0) + \beta(0, -1) = 0$$

$$\Leftrightarrow \begin{cases} \alpha = 0 \\ -\beta = 0 \end{cases} \Rightarrow \text{lin. indep.} \Rightarrow \dim \text{Im } f = 2.$$

(You may use the determinant.)

$$⑤ \quad f \in \text{End}_{\mathbb{R}}(\mathbb{R}^3)$$

$$f(x, y, z) = (-y + 5z, x, y - 5z)$$

Det. a basis and dimension of $\text{Ker } f$, $\text{Im } f$.

$$\text{Ker}(f) = \{ (x, y, z) \in \mathbb{R}^3 \mid (-y + 5z, x, y - 5z) = (0, 0, 0) \}$$

$$\Rightarrow \begin{cases} -y + 5z = 0 \\ x = 0 \\ y - 5z = 0 \end{cases} \Rightarrow y = 5z \Rightarrow \{ (0, 5z, z) \mid z \in \mathbb{R} \} \\
 = z \cdot (0, 5, 1) \Rightarrow \text{basis (lin. ind.)} \\
 \Rightarrow \dim(\text{Ker}(f)) = 1.$$

$$\begin{aligned}
 \text{Im } f &= \{ (-y + 5z, x, y - 5z) \mid f(x, y, z) = (-y + 5z, x, y - 5z) \} \\
 &= \{ (-y, 0, y) + (0, x, 0) + (5z, 0, -5z) \} \\
 &= \langle (-1, 0, 1), (0, 1, 0), (5, 0, -5) \rangle \\
 &\quad \text{but } (5, 0, -5) = 5(1, 0, -1) \Rightarrow \text{not lin. indep.}
 \end{aligned}$$

$\Rightarrow$ a basis for $I_{uf} = \langle (-1, 0, 1), (0, 1, 0) \rangle$

$$\Rightarrow \dim(I_{uf}) = 2.$$

⑦ $A = \{(x, y, z) \in \mathbb{R}^3 \mid x + 2y + 3z = 0\}$ of $\mathbb{R}^3$.

$$B = \{ax + bx^3 \mid a, b \in \mathbb{R}\} \text{ of } \mathbb{R}_3[x].$$

Det. the complement.

$$A = \{(-2y - 3z, y, z) \mid y, z \in \mathbb{R}\}$$

$$= \{y(-2, 1, 0) + z(-3, 0, 1) \mid y, z \in \mathbb{R}\}.$$

$$= \langle (-2, 1, 0), (-3, 0, 1) \rangle.$$

Exam!

$$\dim(\bar{A}) = 1.$$

$$\Rightarrow \bar{A} = \langle (a, b, c) \rangle \text{ s.t. } (a, b, c) \text{ lin. ind. with } (-2, 1, 0), (-3, 0, 1)$$

$$\Rightarrow \begin{vmatrix} -2 & 1 & 0 \\ -3 & 0 & 1 \\ a & b & c \end{vmatrix} = a + 3c + 2b \neq 0$$

Let $a = 1 \quad b = 1 \quad c = 1$

$$\begin{vmatrix} -2 & 1 & 0 \\ -3 & 0 & 1 \end{vmatrix}$$

$$\Rightarrow \bar{A} = \langle (1, 1, 1) \rangle \text{ not unique.}$$

$$B = \langle x, x^3 \rangle$$

$$\Rightarrow \bar{B} = \langle 1, x^2 \rangle \text{ s.t. } f = \alpha_1 \cdot 1 + \alpha_2 \cdot x + \alpha_3 \cdot x^2 + \alpha_4 \cdot x^3$$

⑧ $S = \{(x, y, z) \in \mathbb{R}^3 \mid x = 0\}$

$$T = \langle (0, 1), (1, 1, 0) \rangle$$

$S \cap T = ?$ $\Rightarrow$ not just the set, but also the generated subspace given by it.

Prove $S + T = \mathbb{R}^3$

$$T = \{ \alpha_1 (0, 1, 1) + \alpha_2 (1, 1, 0) \mid \alpha_1, \alpha_2 \in \mathbb{R} \}.$$

$$T = \{ (\alpha_2, \alpha_1 + \alpha_2, \alpha_1) \mid \alpha_1, \alpha_2 \in \mathbb{R} \}.$$

$$S \cap T = \{ (x, y, z) \in \mathbb{R}^3 \mid x=0 \text{ and } y=x+z \}$$

$$= \{ (0, z, z) \mid z \in \mathbb{R} \}.$$

$$= \{ z (0, 1, 1) \mid z \in \mathbb{R} \}$$

$$= \langle (0, 1, 1) \rangle.$$

$$\underbrace{\dim S}_{2} + \underbrace{\dim T}_{2} = \underbrace{\dim(S+T)}_{1} + \underbrace{\dim(S \cap T)}_{1} \Rightarrow \dim(S+T) = 3.$$

$$S = \langle (0, 1, 0), (0, 0, 1) \rangle.$$

$$\underbrace{S, T}_{\text{subsets}} \subseteq \mathbb{R}^3. \Rightarrow S+T \subseteq \mathbb{R}^3.$$

$$\underbrace{\dim(S+T)}_3 \leq \underbrace{\dim \mathbb{R}^3}_3 \Rightarrow S+T = \mathbb{R}^3.$$

(without not even knowing how $S+T$ looks like)